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Monitoring in High Performance Computing

Installing an example monitoring stack (TIG)

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Learning Objectives

- Basic understanding of monitoring
- Basic installation of a monitoring stack

Definition

- Continuous collection of data/metrics from a system
- Analysis of the collected data/metrics within a period of time
- Up-to realtime
- Storing of (selected) data for later analysis

Difference to reporting:

- Monitoring takes place nearly in realtime
- Reporting is about analysing data/metrics over a long period: statistics

Motivation

- W/o monitoring the status of the system/software is unclear
- Admins want monitoring data to check:
 - ▶ Availability
 - ▶ Performance of the system
- Problems, e.g., regressions can be identified using the data/metrics
- Analyze system performance to plan future updates/systems (procurement)
- HPC users - profile/optimize jobs utilizing metrics

What can be monitored?

IT: Availability of network infrastructure/components/services

- PDUs, PSU, CDUs, ...
- Switches, router, servers, compute nodes, ...
- SSH, FTP, Apache web server, ...

Metrics: Performance/Load/Misc.

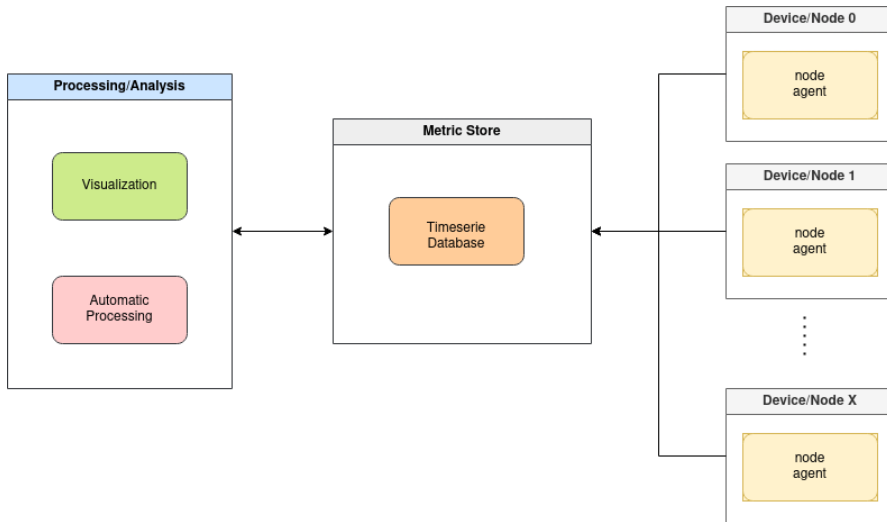
- Infrastructure: bandwidth used, node load, I/O load, storage used, ...
- Job specific: CPU load, bandwidth used, I/O Load, storage/memory used ...
- Energy consumption metrics to be associated later with jobs etc.

Metrics Monitoring infrastructure in HPC

Monitoring stacks usually consists of three components/stages:

- Metrics collector:
 - ▶ collect
 - ▶ aggregate/pre-process
 - ▶ forward/store/provide metrics to database
- Database: collects, stores and provides data (relational, time-series)
 - ▶ collect/stores metrics from collectors
 - ▶ provides the collected data via APIs for further processing
 - ▶ can have retention handling etc.
- Processing: tools accessing the data and the database to
 - ▶ perform automatic data analysis
 - ▶ visual display of data to allow human analysis of the performance/data

Monitoring infrastructure in HPC 2



Components: metrics collector

- task: collect, aggregate/preprocess and provide data to the database
- usually a local process run as service
- usually extendable/utilizing other tools to get metrics
- main approach for providing data:
 - ▶ **pull** - agent waits for the database to ask for the data
 - ▶ **push** - agent pushes collected data automatically to database
- examples: PCP, Telegraf, node-exporter, cc-exporter

Components: metrics database

- follow two main types
 - ▶ relational DB
 - ▶ time-series DB
- offer the typical DB functionality to work in enterprise (fleets, sharding)
- may offer functionality specific to time series data, e.g. data retention, compaction
- examples: InfluxDB, Prometheus, OpenTSDB

Components: metrics database - TSDB

- developed specific for storing time series data
- time series data consists of
 - ▶ time-stamps
 - ▶ tags identifying the metrics/other metadata
 - ▶ data/values
- time series data for a specific measurement are
 - ▶ highly uniform
 - ▶ have no relation/are standalone
 - ▶ easily transferable using text
- uniformity/structure allow for specific optimization, e.g.
 - ▶ compression
 - ▶ transfer rate/ingestion

Components: TSDB vs. relational DB

Relational DB

- Defined table format/schema
- Develop for large amount of data with complex relations
- Fast to process SQL queries on huge sets
- Fast to operate on entries (e.g. modify/delete)
- Difficult to store large amounts of uniform data at the same time

Time-series DB

- Flexible format, no fixed schema
- Developed for storing large amounts of uniform data at the same time
- Fast to store the data
- Slow to operate on data in the DB (e.g., modify/delete entries)

Components: processing

- the collected data can be processed by anything imaginable:
 - ▶ scripts
 - ▶ programmes
 - ▶ visualization software
- use cases, e.g.:
 - ▶ regressions testing (compare system performance before/after changes using)
 - ▶ degeneration - loss of performance due to factors like age, overheating
 - ▶ report generation
 - ▶ analyse program performance
- visualization examples: Grafana, Kibana, Graphite, Cyclotron, matplotlib, ...

TIG Stack

Abbreviation for

- **Telegraf** - the node agent collecting metrics on the systems
- **InfluxDB** - a time-series database used to store the data
- **Grafana** - a tool to visualize metrics and create dashboards

TIG Dashboard example



InfluxDB

- Developed by [InfluxData](#)
- First released 2013
- Written in Go for performance
- Time-series based instead of relational
- Design specifically for time-series from all kind of devices
- Query language Flux (similar to SQL)
- push approach
- open-core

Telegraf

- Developed by [InfluxData](#)
- Node agent for InfluxDB
- Tasks: collecting, processing, aggregating, writing metrics
- Extendable architecture based on plugins
- As InfluxDB written in Go for max. performance
- Is run as service on the monitored systems

Telegraf - Plugins

- Telegraf uses plugins to collect and send data
- the plugins utilize different tools to collect metrics e.g.
 - ▶ Likwid to collect hardware counters from CPUs
 - ▶ Files in /procfs for process statistics
- 2 plugin types: input and output
- At least one input and one output plugin has to be configured e.g.
 - ▶ Input: CPU, diskio, procstat
 - ▶ Output: influxdbv2
- Huge variety of plugins can be found the [Plugin documentation](#)

Grafana

- Developed by [GrafanaLabs](#)
- Web based dashboard application to visualize data
- Written in Typescript, Go
- build in webserver based on Go http server
- Supports relational, time-series databases
- open-core
- Can display real-time data using specific plugins/config (MQTT+Telegraf)
- Goals: easy visual analysis of data from different sources, easy dashboard creation

Installation of TIG

- Tutorial 1
- Tutorial 2
- Tutorial 3
- Tutorial x

Plenary discussion

- issues
- impression/experience
- questions/open points?

References

- <https://www.influxdata.com/>
- <https://grafana.com/>
- <https://docs.influxdata.com/telegraf/v1.20/plugins/>